Flexible Drives RAP

Due 23 Sep at 16:30 **Points** 12 **Questions** 12
Available 23 Sep at 16:00 - 23 Sep at 16:30 30 minutes **Time limit** 20 Minutes

Instructions

This is a CLOSED BOOK multiple choice quiz. There are 12 questions to be completed.

The quiz is to be completed individually. No communication with others is allowed at any time.

Choice the best answer for the question.

Time limit: 20 minutes.

By starting the quiz you agree to abide by these rules.

This quiz was locked 23 Sep at 16:30.

Attempt history

	Attempt	Time	Score
LATEST	Attempt 1	20 minutes	10 out of 12

ⓘ Correct answers are hidden.

Score for this quiz: **10** out of 12
Submitted 22 Sep at 22:55
This attempt took 20 minutes.

Incorrect

Question 1

0 / 1 pts

On the topic of pitch diameter (PD), which of the following statements is TRUE:

- Pitch diameter is a measure of the length of a link in a chain

The pitch diameter for a timing belt is slightly larger than the pulley diameter

None of these statements are true

It is not necessary to know the pitch diameter in order to calculate the force in a V-belt

The pitch diameter of chain sprocket is the maximum diameter

Incorrect

Question 2

0 / 1 pts

In terms of Timing Belts, which statement is FALSE:

Chordal speed variation is not a consideration

Lubrication is not a requirement for smooth operation

The force transfer between sprocket and belt takes place over the first 6 teeth of engagement

A high level of pre-tension is required

Can never be used with idlers

Question 3

1 / 1 pts

A timing belt is functionally most similar to which of the following power transfer mechanisms:

- [x] roller chain
- [ ] flat belt
- [ ] all 3 of the drives mentioned
- [ ] none of the three drives mentioned
- [ ] V-belt

Question 4

1 / 1 pts

Which of the following statements about roller chains is TRUE?

- [ ] Lubrication provides only a small increase in chain life
- [ ] Velocity ratios of around 8:1 are ideal
- [ ] They tend to fail due to lack of tensile strength
- [ ] The chain speed always remains constant if the sprocket has uniform angular velocity
- [x] They tend to fail due to wear

Question 5

1 / 1 pts

The recommended minimum number teeth for a chain sprocket is:

- [x] 17
- [ ] 20

- 15
- 12
- 24

Question 6

1 / 1 pts

When using wire ropes to hoist items overhead using a crane or winch, a minimum safety factor should be at least:

- 3
- 1.5
- 5
- 18
- 12

Question 7

1 / 1 pts

To convert from the power units of ft-lbs/sec to horsepower use the following conversion

- Divide by 32.2
- Divide by 33000
- Divide by 550
- Multiply by 550

- Divide by 454

Question 8

1 / 1 pts

For a flat belt drive, the maximum transmitted power is LEAST sensitive to SMALL changes in which of the following:

- wrap angle on the driving pulley
- initial tension in the belt
- belt width
- belt speed
- angle between the shafts of the driving and driven pulleys

Question 9

1 / 1 pts

The term *“Pre-Extreme Horespower”* refers to the following:

- The highest degree of lubrication
- The power rating for a chain where link fatigue failure is the primary mode of failure
- The maximum horsepower rating for a chain
- The breaking strength of a synchronous belt

The power rating for a chain where roller fatigue or excessive wear are the modes of failure

Question 10

1 / 1 pts

Which of the following statements is FALSE?

- Lubrication should never be used for wire ropes
- The breaking strength of the wire rope is a function of the number of wires in the rope and the material strength of the wire
- Wire ropes are used for lifting, moving and hauling
- Wire ropes should never be tied in knots
- The term Lay refers to the direction of the wires in the strand and the direction of wrap for the strands

Question 11

1 / 1 pts

When specifying the working load of wire rope, which of the statements is FALSE?

- one must consider loads caused by sudden stops and starts
- one must only consider the maximum sheave diameter

- one must take in account the maximum allowable bearing pressure of the wire rope on the sheave
- one must consider the modulus of elasticity of the wire rope
- one must consider the application in order to specify a safety factor

Question 12

1 / 1 pts

A variable-speed belt drive can be designed by:

- changing the tension in the belt to increase friction, decreasing the drive ratio
- **changing the effective driving and driven pulley diameters by using conical pulleys**
- changing the drive from toothed belts to V-belts
- changing the relative angle of the pulley shafts through 90° to double the drive ratio
- increasing the distance between pulleys, stretching the belt to increase the drive ratio

Quiz score: **10** out of 12

Bearings and Bushings RAP Quiz

Due 18 Nov at 23:59 **Points** 12 **Questions** 12
Available 18 Nov at 7:00 - 18 Nov at 23:59 16 hours and 59 minutes
Time limit 20 Minutes

Instructions

This is a CLOSED BOOK multiple choice quiz. There are 12 questions to be completed.

The quiz is to be completed individually. No communication with others is allowed.

Choose the best answer for the question.

Time limit: 20 minutes.

By starting the quiz you agree to abide by these rules.

This quiz was locked 18 Nov at 23:59.

Attempt history

	Attempt	Time	Score
LATEST	Attempt 1	20 minutes	9 out of 12

ⓘ Correct answers are hidden.

Score for this quiz: **9** out of 12
Submitted 18 Nov at 17:57
This attempt took 20 minutes.

Question 1 — 1 / 1 pts

The term dynamic or absolute viscosity is defined as:

- The ability of a fluid to resist density changes
- The size and shape of the boundary layer profile

- The relationship that states the shear stress gradient is proportional to temperature
- The relationship between surface tension and viscous forces
- The relationship that states the shear stress is proportional to velocity gradient

Question 2

1 / 1 pts

Which of the following is NOT a characteristic of boundary lubricated bearings (bushings)?

- The PV rating is derived from the maximum pressure and linear speed.
- Temperature is a consideration in material selection.
- Bushings often have oils embedded in the material.
- Materials used for bushings are plastics and bronzes.
- The maximum pressure is a design criteria.

Question 3

1 / 1 pts

There is a trade-off between the load on a bearing (F) and its useful life (L), most simply expressed as:

- F / L = constant

- (selected) FL^1/a = constant, where a depends on the type of rolling element.
- Manufacturers of bearings all develop their own individual measures of performance.
- F L = constant
- F^L = constant

Question 4

1 / 1 pts

The L_{10} life (also called rating life) is given by the number of revolutions that:

- (selected) 90% of bearings will achieve or exceed before failing.
- An average bearing will achieve under a load 10% greater than the catalog load rating.
- An average bearing will achieve under a load 10% less than the catalog load rating.
- 50% of bearings will achieve or exceed before failing.
- 10% of bearings will achieve or exceed before failing.

Question 5

1 / 1 pts

How are thrust loads included in ball bearing selection?

The sum of the radial and thrust loads is used (i.e. Fcombined = Fradial+Fthrust).

The equivalent radial load that does the same damage as thrust and radial loads combined together.

The larger load (radial or thrust) is used.

The thrust load can always be ignored and the radial load is used.

The radial load can always be ignored and the thrust load is used.

Incorrect

Question 6

0 / 1 pts

When fitting a bearing on a shaft that supports a chain drive, how should the bearing be mounted?

Both inner and outer races should be a clearance fits.

The inner race of the bearing should be a clearance fit and the outer race be an interference fit.

Both inner and outer races should be interference fits.

It does not matter about the fits.

The inner race should be an interference fit and the outer race be a clearance fit.

Question 7

1 / 1 pts

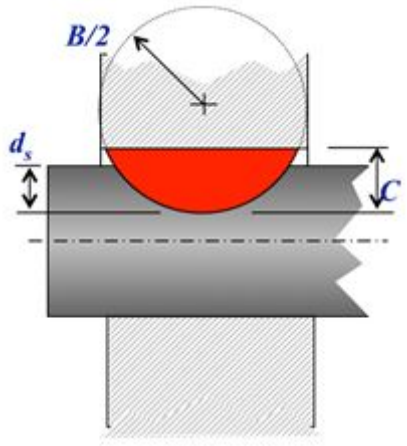
One of the drawbacks of using retaining rings on a shaft is that they:

- (selected) can cause high stress concentrations.
- require a shaft shoulder to be machined in addition to a groove.
- provide low transfer of thrust (axial force).
- do not allow for precise axial positioning of components.
- can be easily removed.

Question 8

1 / 1 pts

In the image below, what type of component is depicted attaching the hub with the shaft?



- (selected) Woodruff key
- Round pin
- Gib-head key
- Retaining ring

Square key

Question 9

1 / 1 pts

Tapered roller bearings are used in pairs principally:

- To increase the maximum allowable speed of the shaft.
- To increase bearing life.
- To facilitate the mounting of the bearings on the shaft.
- To allow backlash to be easily adjusted.
- (selected) To constrain the axial direction and produce axial preload.

Incorrect

Question 10

0 / 1 pts

Which of the following is an important design consideration for a component being located on a shaft by a square or rectangular key?

- (selected) The key must be inserted in the keyway with a mechanical interference fit.
- The steel chosen should be harder than the shaft steel.
- A set screw should never be used in conjunction with a key to avoid damage to the key surface.

The key should be longer than the component to reduce stress concentrations.

The key must be constrained or secured to prevent axial motion of the key and component.

Question 11

1 / 1 pts

In terms of shaft design, which of the following statements is FALSE?

The diameter of the shaft, at the location of a chain sprocket, will involve both the torque and bending moment.

Stress concentration factors are only applied to the alternating bending stress term in the ASME (Mott) design equation.

The stress concentration factor used for fatigue is the same as the stress concentration factor used for static loads

For a bearing located on the end of shaft, the shaft diameter calculation will involve only reversing vertical shear.

For most applications, only two bearings should be used to support a shaft .

Incorrect

Question 12

0 / 1 pts

Which of the following cannot be used to transmit torque between elements mounted on shafts?

- a Woodruff key
- a shear pin
- a retaining ring
- a spline
- a press fit

Quiz score: **9** out of 12

Springs and Fasteners RAP Quiz

Due 1 Dec at 15:30 **Points** 12 **Questions** 12
Available 1 Dec at 14:00 - 1 Dec at 15:30 1 hour and 30 minutes **Time limit** 20 Minutes

Instructions

This is a CLOSED BOOK multiple choice quiz. There are 12 questions to be completed.

The quiz is to be completed individually. No communication with others is allowed.

Choose the best answer for the question.

Time limit: 20 minutes.

By starting the quiz you agree to abide by these rules.

This quiz was locked 1 Dec at 15:30.

Attempt history

	Attempt	Time	Score
LATEST	Attempt 1	14 minutes	12 out of 12

ⓘ Correct answers are hidden.

Score for this quiz: **12** out of 12
Submitted 30 Nov at 10:47
This attempt took 14 minutes.

Question 1 — 1 / 1 pts

The standard bolt/screw sizes in both imperial and metric systems are catalogued by:

- ○ Root diameter
- ◉ Major diameter

- Tensile stress area
- Clearance hole diameter
- Screw length

Question 2

1 / 1 pts

For Ball Screws, which of the following statements is FALSE:

- The lifetime of ball screws are defined by the length of travel that the screw makes for a given load
- Column buckling is not a consideration in selecting a ball screw
- Ball screws have efficiencies of approximately 90%
- The relationship between the ball screw life and load is the same form as used for ball bearings
- Ball screws are never self-locking

Question 3

1 / 1 pts

What is the name of the fastener shown?



- (●) Socket head cap screw
- ( ) Round head machine screw
- ( ) Flat head cap screw
- ( ) Carriage bolt
- ( ) Lag bolt

Question 4

1 / 1 pts

Which of the following does NOT directly influence the spring rate of a bolt in a tension joint?

- ( ) The length of the threaded portion of the grip
- (●) The pitch of the threads
- ( ) The Youngs modulus of the bolt material
- ( ) The length of the unthreaded portion of the grip
- ( ) The tensile stress area

Question 5

1 / 1 pts

Considering fatigue loading of bolted tension joints, which of the following statements is TRUE?

- The endurance strength increases as SAE bolt grade (or metric property class) decreases
- Fatigue stress concentration factors can be ignored
- Endurance strength of rolled threads is higher than cut threads
- For fatigue loading, cut threads and rolled thread are the same
- Fatigue failures always occur on the unthreaded portion of the bolt

Question 6

1 / 1 pts

Where is the rotational pivot point for a shear joint held by multiple bolts?

- The centroid of the cross-sectional areas of the bolts
- The center of the middle-most bolt
- The average x- and y-coordinates of all bolts
- The center of the bolt located at the minimum distance perpendicular to the load
- The center of the bolt located at the maximum distance perpendicular to the load

Question 7

1 / 1 pts

What size wrench is needed to tighten a 1/2 inch nut?

- 1"
- 9/16"
- 1/2"
- 3/4"
- 5/8"

Question 8

1 / 1 pts

The spring rate of a compression spring:

- determines sensitivity to cyclic loading
- is a measure of length over deflection
- is a measure of force over spring length
- is a measure of force over deflection
- is a measure of the curvature effect

Question 9

1 / 1 pts

For a compression or extension spring, the index C is a measure of coil curvature and relates:

- strength of actual spring material over strength of spring steel in shear
- coil diameter and spring constant K
- diameter of wire over length
- diameter of coil over length
- coil diameter to wire diameter

Question 10

1 / 1 pts

In analyzing a helical compression spring, the wire is considered to be subjected to:

- bending stress
- direct shear only
- normal compression stress
- tension stress
- direct shear and torsional shear

Question 11

1 / 1 pts

The curvature effect of coiled compression springs (factors K_B and K_W) causes:

- reduced stress at the ends of the spring
- increased deflection due to buckling in long springs
- increased stress at the ends of the spring
- increased stress in springs made of materials other than steel
- unequal stress between the inside and outside portions of the spring coil

Question 12

1 / 1 pts

A metric bolt with the ISO grade designation of 4.8 has following approximate material strengths:

- 800 MPa Ultimate Strength and 400 MPa Yield Strength
- 400 MPa Ultimate Strength and 280 MPa Yield Strength
- 520 MPa Ultimate Strength and 415 MPa Yield Strength
- 480 MPa Ultimate Strength and 380 MPa Yield Strength
- 400 MPa Ultimate Strength and 320 MPa Yield Strength

Quiz score: **12** out of 12

Gear Module RAP Quiz

Started: 11 Oct at 21:48

Quiz instructions

This is a CLOSED BOOK multiple choice quiz. There are 12 questions to be completed.

The quiz is to be completed individually. No communication with others is allowed while taking the quiz. No information regarding the contents of the quiz is to be shared with others.

Choose the best answer for the question.

Time limit: 20 minutes.

By starting the quiz you agree to abide by these rules.

Question 1

1 pts

Which of the following gears has one gear with teeth wrapped around the shaft, resembling a screw, meshing with another gear?

- ○ Helical gear
- ◉ Worm gear
- ○ Spur gear
- ○ Hypoid gear
- ○ Bevel gear

Question 2

1 pts

Which of the following statements is FALSE?

- ○ Two right-handed helical gears with 45° helix angle can work on perpendicular shafts
- ◉ When a pinion gear meshes with an internal gear, their shafts rotate in opposite directions

- ○ The number of teeth on any gear must be an integer
- ○ Bevel, worm, and mitre gears work on perpendicular shafts
- ○ Spur and helical gears operate on parallel shafts

Question 3

1 pts

For spur gears, which parameter is the reciprocal of the diametral pitch?

- ○ Dedendum
- ○ Clearance diameter
- ○ Pitch diameter
- ○ Addendum
- ◉ Module

Question 4

1 pts

For two mating gears, which one of the following has to be the same for both gears?

- ○ Addendum diameter
- ○ Base circle diameter
- ○ Gear width
- ◉ Diametral pitch
- ○ Angular velocity

Question 5

1 pts

Why is the involute profile an important aspect of gear tooth geometry?

- (●) It ensures constant angular-velocity ratio between two gears
- ( ) It provides the largest possible stiffness for a given tooth volume
- ( ) It ensures that mechanical interference will never occur
- ( ) It ensures that the gear pitch circles of the two gears are tangent to each other
- ( ) It is the only gear profile that can be manufactured by hobbing

Question 6

1 pts

For this question, consider a well-designed gear set (spur or helical) transmitting a torque from a pinion to a driven gear.

When calculating forces, the tangential loading between the gears predominantly depends on:

- ( ) Helix angle
- ( ) Face width
- (●) Pressure angle
- ( ) Mechanical interference
- ( ) Involute profile

Question 7

1 pts

The surface pressure on the face of a gear tooth is proportional to which of the following?

- (●) Transmitted torque
- ( ) Heat treatment of the steel

- ○ Mechanical interference
- ○ Pressure angle
- ○ Helix angle

Question 8

1 pts

One pound mass is equivalent to:

- ◉ 0.454 kg
- ○ 746 g
- ○ 4.905 N
- ○ 3.78 kg
- ○ 2.2 lbf

Question 9

1 pts

Which of the following statements is *always* TRUE for a planetary gear train with multiple planet gears? (assume the planetary gear train is not connected to anything else)?

- ◉ The planetary gear train has more degrees of freedom than a compound gear train
- ○ Only one of the planetary gears may mesh with the sun gear
- ○ The planetary gears can never provide a reverse rotation
- ○ All of the planetary gears must be of different diameters from each other
- ○ There must always be an even number of planetary gears

Question 10

1 pts

For a 3-gear drivetrain, for which the middle gear is in mesh with an input (driving) gear on one side and an output (driven) gear on the other, which one of these statements is true about the overall drive train value *e* ?

- ○ |e| < 1 when the output gear has more teeth than the input gear
- ○ e is greater for helical gears than spur gears, with gears of equal number of teeth
- ◉ |e| < 1 when the output gear has fewer teeth than the input gear
- ○ e is the ratio of the width of the driven gear over the width of the driving gear
- ○ e depends on the number of teeth of the middle gear

Question 11

1 pts

A rack can be used in shaping or cutting gears. What property of a rack is especially convenient?

- ○ The sides of an involute rack are straight and therefore accurate
- ○ Cutting gears with a rack is the fastest gear forming method for a given gear size
- ◉ At least TWO of the conditions are true
- ○ The same rack can machine gears of different diametral pitch
- ○ A rack is linear as opposed to circular

Question 12

1 pts

Which of the following statements is FALSE for a wormgear drive

- ○ Efficiencies of wormgears are less than bevel gears
- ◉ The axial velocity of the worm is the tangential velocity of the gear
- ○ High gear ratios can be achieved using worm gears
- ○ The efficiency of the wormgear is a function of the friction coefficient and the pressure angle
- ○ For wormgears to be self-locking, the lead angle must be kept low (no more than 5°)

Saved at 22:06 Submit quiz

Fluid Power RAP quiz

Due 31 Oct at 23:59 **Points** 12 **Questions** 12
Available 31 Oct at 7:00 - 31 Oct at 23:59 16 hours and 59 minutes
Time limit 20 Minutes

Instructions

This is a CLOSED BOOK multiple choice quiz. There are 12 questions to be completed.

The quiz is to be completed individually. No communication with others is allowed.

Choice the best answer for the question.

Time limit: 20 minutes.

By starting the quiz you agree to abide by these rules.

This quiz was locked 31 Oct at 23:59.

Attempt history

	Attempt	Time	Score
LATEST	Attempt 1	20 minutes	10 out of 12

ⓘ Correct answers are hidden.

Score for this quiz: **10** out of 12
Submitted 31 Oct at 22:28
This attempt took 20 minutes.

Question 1 — 1 / 1 pts

Liquid enters and is forced out of a hydraulic pump by?

- (●) Generating an increasing and decreasing volume within its housing
- ( ) Absorbing additional energy from its prime mover

- None of the these answers
- Generating centrifugal forces that form a positive seal between vane and housing
- Overcoming any resistance to flow

Question 2

1 / 1 pts

1. The prime function of an accumulator is to?

- Serve a relief tank
- Store potential energy for a hydraulic system
- Apply whatever pressure is needed to push the actuator
- Deliver a continuous source of flow for a hydraulic system
- Convert hydraulic working energy to a mechanical force

Question 3

1 / 1 pts

A _____ allows flow to bypass a flow control valve:

- Relief valve
- Directional valve
- Pressure compensated flow control valve
- Check valve

Needle valve

Question 4

1 / 1 pts

With regards to directional valves, which of these statements is FALSE?

Directional valves can be operated by mechanical, manual, electrical, or pilot pressure means

4-way directional valves allow actuators or hydraulic motors to reciprocate (go back and forth)

Directional valves can be a 4-way, 3-way, or 2-way type

The moving part that directs the flow in the valve is called the “spool”

Springs should never be used in a directional valve

Question 5

1 / 1 pts

With regards to pressure control valves, which of these statements is FALSE?

Simple pressure control valves have their movable parts biased (opposed) by spring pressure only

An unloading valve is a remotely operated normally closed pressure control valve that directs flow to the tank when the pressure in a remote part of the system reaches a predetermined pressure level

A sequence valve is a normally closed pressure valve which causes one operation to occur before another operation by blocking fluid passages until a set pressure is achieved

“Cracking pressure” refers to the maximum pressure the valve can withstand before it bursts

A relief valve is a normally closed valve which diverts flow to the reservoir at a pre-determined pressure level

Question 6

1 / 1 pts

Flow rate in a hydraulic system is expressed as (imperial units)?

RPM

SCFM

GPM

PSIA

FPS

Question 7

1 / 1 pts

With regards to fluid connectors, which of these statements is FALSE?

- Pipe connections are made by threaded fittings while tube fittings are made by flared, or flareless, tube fittings
- Pipe is a rigid conductor which is not intended to be bent or shaped into a configuration
- Pipe size is indicated by a nominal designation for the outside diameter with the wall thickness identified by a schedule number
- (selected) Tubing runs can only be assembled as a straight line
- A hose should be used whenever a hinge joint or other flexible joint is used on a machine.

Question 8

1 / 1 pts

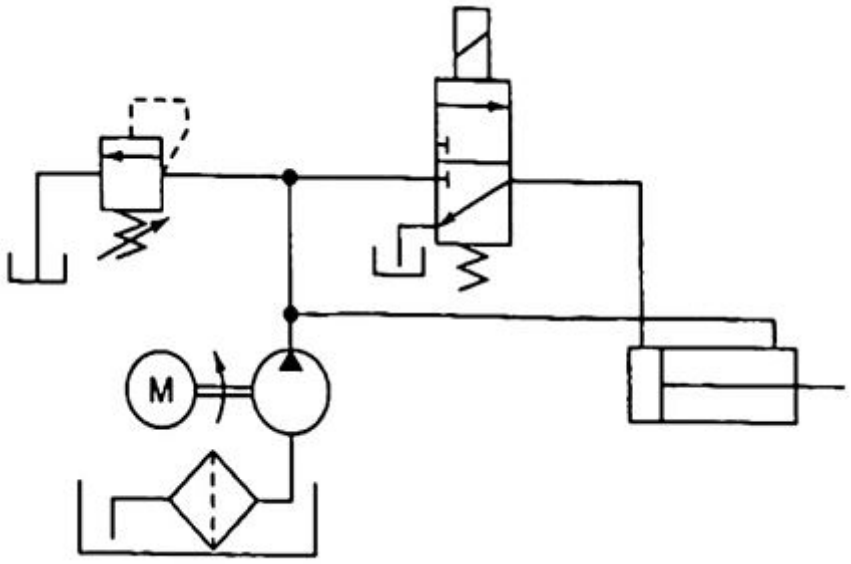
A positive displacement pump applies a slightly higher pressure to the fluid than required at the actuator because of?

- none of the factors
- (selected) all three factors
- fluid changing direction
- viscosity
- fluid friction due to restrictions

Incorrect

Question 9

0 / 1 pts

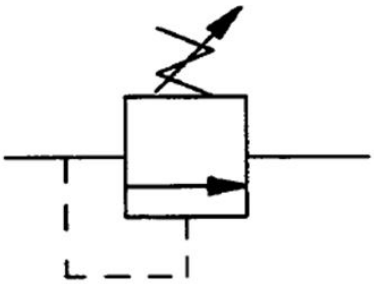


With respect to the above figure, which of the following statements is FALSE?

- This is an example of a regenerative circuit
- The pressure is equal on both ends of the actuator
- Engaging the directional valve will cause the actuator to extend
- The maximum working pressure of the system is set by the relief valve
- This circuit means the pump will not output a constant volume of flow

Question 10

1 / 1 pts



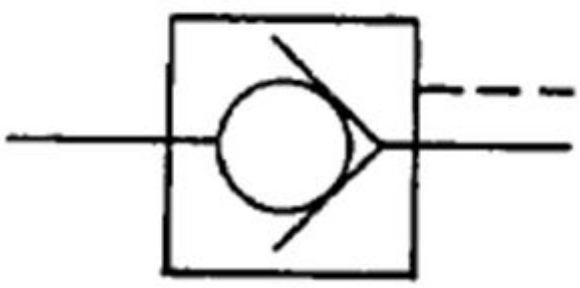
The following symbol refers to?

- Unloading valve
- Pressure reducing valve
- Temperature compensated pressure valve
- Normally open pressure control valve
- Normally closed pressure control valve

Incorrect

Question 11

0 / 1 pts



The following symbol refers to:

- Check valve, pilot operated to close
- Check valve

- Pilot operated flow control valve
- Check valve, pilot operated to open
- Pressure compensated control valve

Question 12

1 / 1 pts

Which of the following statements is FALSE for hydraulic pumps?

- Spur, helical and herringbone are examples of gear pumps
- Piston pumps generate a pumping action by causing pistons to reciprocate using a swash plate
- A gerator design is an example of an internal gear pump
- Vane pumps can be variable volume and/or pressure compensated
- Gear pumps output a volumetric flow rate that varies with the pressure

Quiz score: **10** out of 12